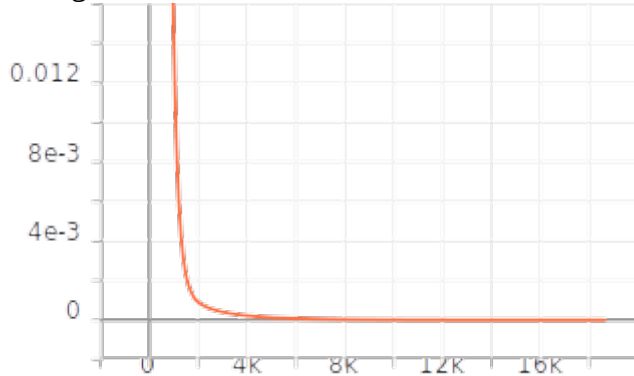


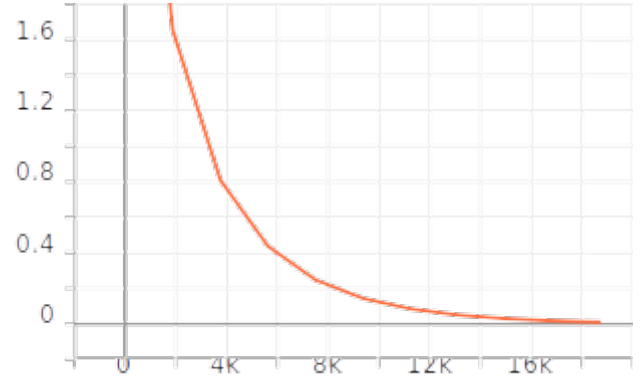
Task 1 Self supervised pre-training

Trained for 10 epochs and got besst accuracy.

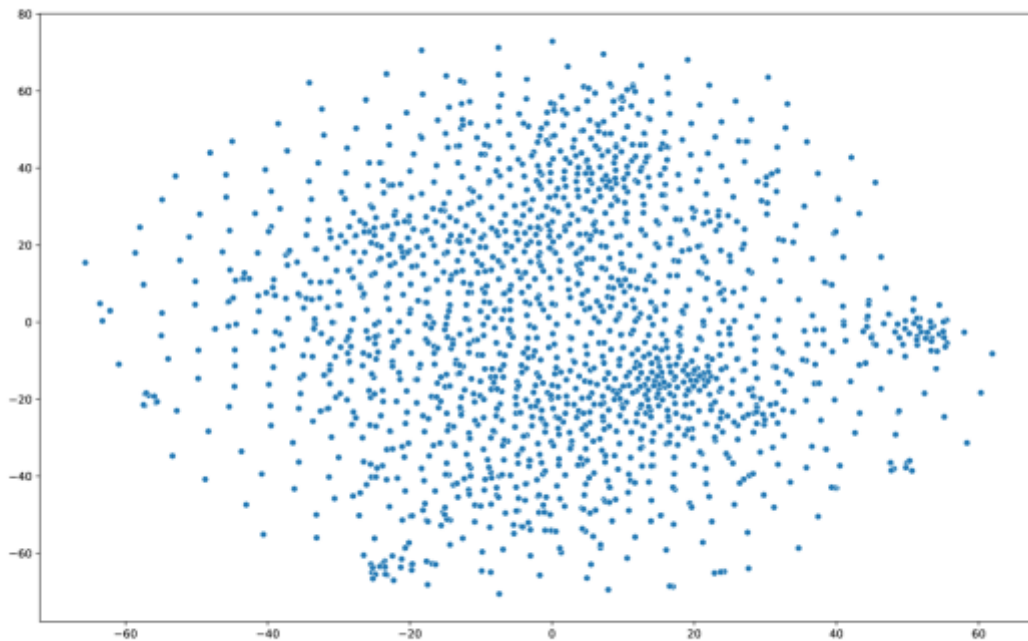
Training Loss



Validation Losss



Quality of embeddings:
TSNE embedding at epoch 0.



TSNE embedding at epoch 9

In the 0th epoch all the points in the center are closer to each other representing all type of images have less distance in the embbding space. Where as in case of 9th epoch it has become little spreaded and groped in distanct places.

It very hard to come to conclusions just by seeing a 2 dimensional embeddings. So the best way to qualitatively analyse the emmbedding is by running the nearest neighbour algorithm on it.

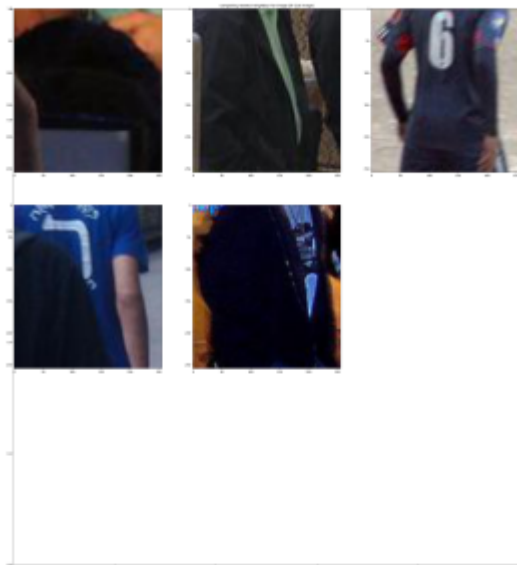
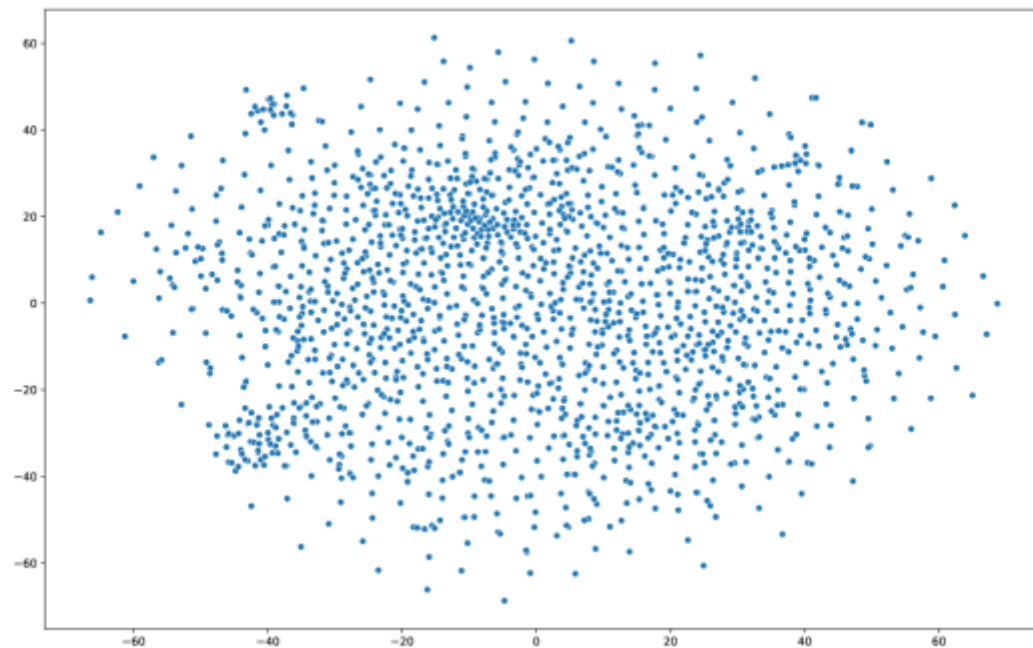
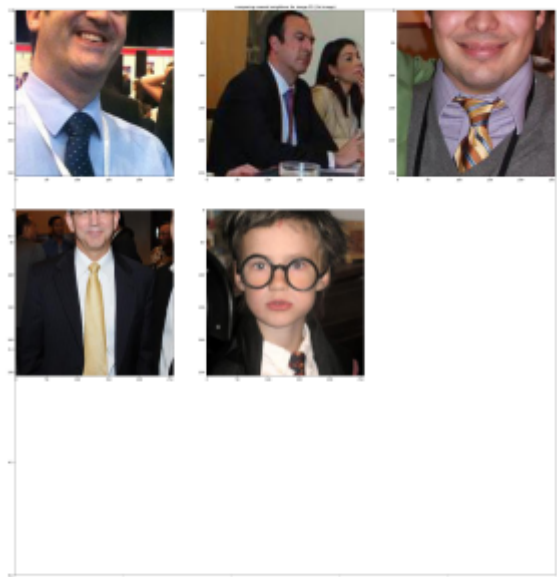


image 1

image 2



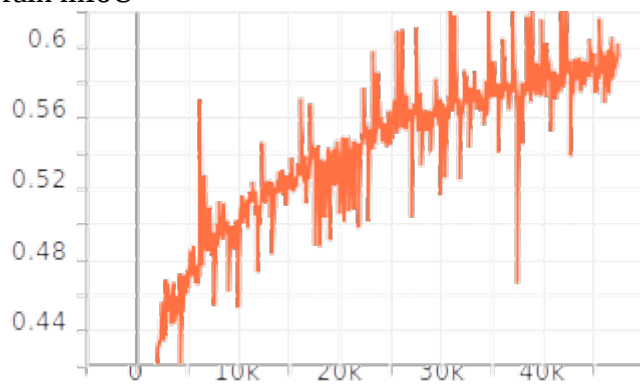
Completed the nearest neighbour algorithm and got the above results. As we can observe clearly from the image 1 all the images have same color blue and in image 2, it focused more on face and lips of the person. Hence we can confirm that DINO has learned a good representation.

Task 2. Attention-based Semantic Segmentation

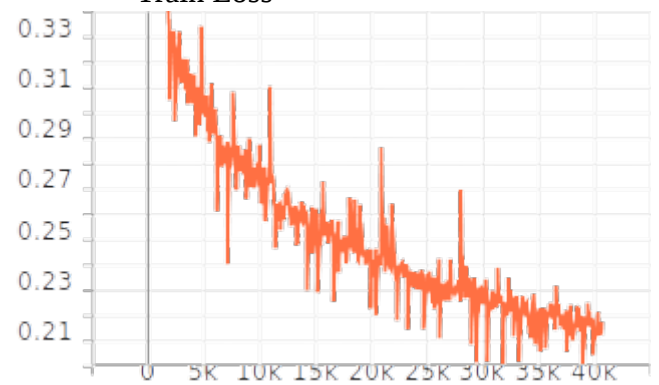
Trained the network for 10 epochs with pretrained weights given from the assignment. Obtained best mIoU of 0.2010.

Trained the network for 10 epochs with pretrained weights from DINO (i.e. task1). Obtained the best mIoU of 0.24

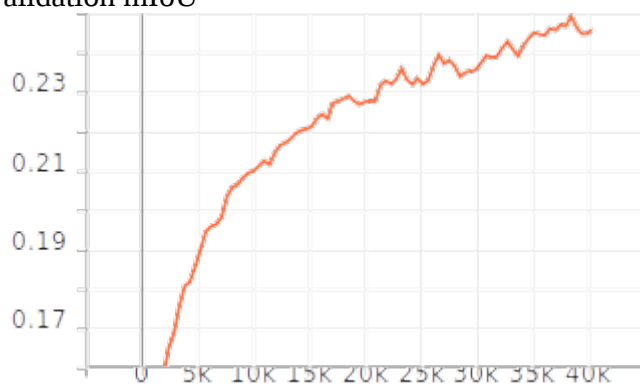
Train mIoU



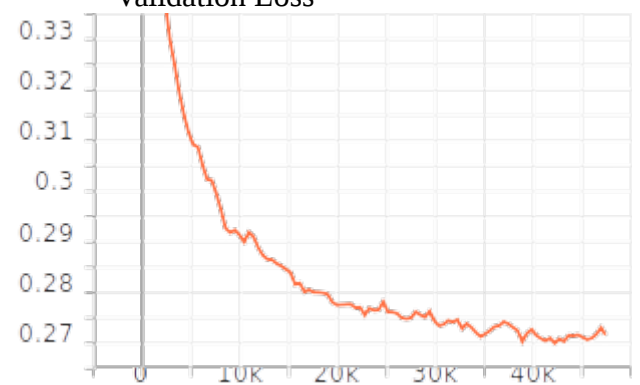
Train Loss



Validation mIoU



Validation Loss



Trained the network for 100 epochs got mIoU of 0.21 in the 10th epoch but took lot of time to reach the mIoU of 0.24.